

Optimizing Distribution and Capacity Expansion at Anadolu Efes

MBAn 558 – Prescriptive Analytics
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Optimization Framework & Network Structure

Short-Term (Linear Programming Model)

Objective: Minimize transportation cost

Decision variables:

- Malt shipment quantities from malt plant i to brewery j
- Beer shipment quantities from brewery j to distribution center k

Constraints:

- Malt plant capacity limits
- Brewery production capacity
- Malt-to-beer yield conversion
- Demand satisfaction at all distribution centers

Long-Term (Mixed Integer Programming Model)

Objective: Minimize Transportation + Fixed Investment

Additional binary decisions:

- Open a brewery at potential site (0/1)
- Expand a brewery (0/1)

Additional constraints:

- Capacity available only if site is opened
- Expansion allowed only if opened
- All supply-demand balance constraints maintained

Efes Supply Chain Network Structure



Improving the Current Distribution Plan

The optimized LP solution^[A1] reallocates malt and beer flows across the network to reduce total transportation cost.

Key Shipment Adjustments

- Malt flows are reallocated more heavily toward cost-efficient brewery routes.
- Beer shipments are redistributed across distribution centers to minimize shipping distance and cost.
- Some previously used routes are eliminated where transportation costs are relatively high.
- Export shipments are optimized to reduce unnecessary cross-network routing.

Binding Constraints & Network Behavior

- Brewery capacity constraints (Istanbul and Ankara) are binding or near-binding.
- Malt plant capacity is fully utilized at cost-efficient routes.
- Demand constraints are strictly satisfied (no unmet demand).

The optimized LP solution reduces total transportation cost to: **\$11.80 million**. This represents a **~16.7% reduction** relative to the current distribution plan.

The current distribution system is not cost-minimizing. Implementing the LP-optimized shipment plan can generate substantial recurring cost savings with no capital expenditure. Immediate implementation is recommended.

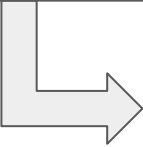
Is Importing Malt Cost-Effective?

From our LP solution sensitivity analysis ^[B-4], we can see that import routes are not cost-effective, they increase total costs.

The Cost Reduction Required for Importing to Become Viable

Route	Current Cost	Required Decrease	Break-Even Cost
Import → Istanbul	0.03178	0.002884	≤ 0.02889 (~9% ↓)
Import → Ankara	0.03251	0.014257	≤ 0.01826 (~44% ↓)

The required decrease values indicate the minimum decrease required in the import cost for imported malt to enter the optimal solution. For shipments to Istanbul, the unit cost must decline by approximately 9%, while shipments to Ankara would require a substantially larger reduction of about 44%. *These thresholds demonstrate that, under current parameters, domestic malt supply remains significantly more cost-efficient than imported alternatives.*



Importing malt could become economically attractive if domestic malt capacity becomes binding, if domestic transportation costs increase materially, or if future demand exceeds domestic supply capability. In such scenarios, imported malt would serve as a marginal supply option.

Impact of Demand Variation on Transportation Cost

We can use the sensitivity analysis ^[B-4] to evaluate how demand fluctuations affect total transportation cost and routing structure.

If actual demand is below allowable range

- Some active shipment routes may drop to zero.
- Previously binding capacity constraints become slack.
- Production may become concentrated at fewer breweries.
- Shipment pattern may restructure to eliminate higher-cost flows.



Demand drops beyond allowable ranges may deactivate routes and create underutilized capacity. Although total transportation cost declines, operational efficiency and average cost performance may deteriorate.

When actual demand exceeds allowable range

- Binding brewery capacity constraints tighten further.
- Previously unused shipment routes may enter the solution.
- Higher-cost transportation links may become active.
- Production may need to shift across breweries.



Large demand increases can exceed sensitivity limits, activate higher-cost routes, and tighten capacity constraints. Transportation costs may rise disproportionately, indicating the need for structural adjustments or capacity expansion.

Optimal Brewery Location and Expansion Strategy

We used the mixed-integer optimization model, [\[63\]](#) to select a network configuration that minimizes total system cost (transportation + fixed investment) while satisfying projected future demand.

The Cost Minimizing Solution:

- **Open new breweries in Izmir, Samsun, and AnkNew**
- **Implement expansion of the new Izmir brewery**
- Not open in Sakarya, Adana, or IstNew

This configuration results in a total system cost of **\$23.56 million**, including **\$12.0 million in yearly equivalent fixed investment costs**.

Rationale and Network Impact

The opening and expansion decisions are driven by geographic demand concentration and transportation cost tradeoffs.

- **Izmir** is both opened and expanded because it efficiently serves a high-demand western region and supports export flows through the Izmir harbor, significantly reducing long-haul shipments from Istanbul and Ankara.
- **Samsun** is opened to directly serve northern demand, eliminating costly cross-country transportation from central breweries.
- **AnkNew** is selected to reduce shipment distances to central and eastern regions (e.g., Kayseri), relieving pressure on the existing Ankara facility.

The optimal solution indicates that targeted regional expansion, rather than broad facility rollout, minimizes total cost. Opening only the most strategically located breweries achieves cost efficiency while avoiding unnecessary capital expenditure.

Non-Optimal Scenarios and Policy Tradeoffs

To understand the managerial tradeoffs, we evaluated two non-optimal scenarios, ^(D4)~~(E5)~~ and compared their total system costs to the optimal configuration

Scenario 1: Force Adana Open, No Samsun

Under this policy, Adana is opened while Samsun is not selected. This increases fixed cost from **\$12.0M to \$13.0M**, but more importantly, total system cost rises to **\$27.06M**, an increase of approximately **\$3.50M (+14.9%)** relative to the optimal solution.

Opening Adana instead of Samsun increases total cost the most because the forced location choice reduces geographic fit with demand and raises recurring transportation costs enough to outweigh the small fixed-cost differences. In other words, Adana's strategic value would need to come from non-cost benefits (e.g., commercial presence in the south, resilience, political/regulatory considerations), because it is not justified on cost alone.

Scenario 2: Do Not Open or Expand Izmir

In this scenario, Izmir is neither opened nor expanded. Although this reduces fixed investment to **\$10.0M**, total system cost increases to **\$26.09M**, which is approximately **\$2.53M (+10.7%)** higher than the optimal solution.

No Izmir site is informative because it shows that *lower* fixed cost does not necessarily mean *lower* total cost. Even though fixed cost drops to 10.0, total cost increases materially, implying that Izmir's location (and expansion) delivers large transportation savings; likely by serving western demand and supporting export flows efficiently. Management would only accept this policy if avoiding Izmir has strong external benefits (e.g., site feasibility, water constraints, execution risk) that exceed the ~\$2.53M annual cost penalty.

Effect of Operating at a Lower Capacity

To evaluate operational flexibility, brewery capacity constraints were reduced to 90% of full capacity, and the model was re-optimized.^[F6]

Under this constraint, total system cost increases from **\$23.56M (full capacity)** to **\$25.51M**, an approximately **+8.3%**. Fixed investment cost rises from **\$12.0M to \$13.8M**, as additional expansion decisions are triggered to compensate for reduced utilization limits.

Reducing capacity forces the system to rely more heavily on additional facilities and expansions to satisfy demand. While transportation patterns adjust accordingly, the network becomes more capital-intensive to maintain feasibility.

However, operating below full capacity provides operational advantages. Lower utilization reduces production strain, allows maintenance flexibility, improves service reliability under demand uncertainty, and reduces the risk of supply disruptions. From a strategic perspective, the modest cost increase may be justified by improved resilience and operational stability.

The analysis shows that while full-capacity operation is cost-minimizing, maintaining a buffer (e.g., 90% utilization) increases total cost moderately but enhances flexibility and risk mitigation.

Final Recommendations

Short-Term Actions (0–12 Months)

Implement the LP-optimized distribution plan:

The short-term model reduces transportation cost to \$11.80M, generating approximately 16.7% savings with no capital investment required

Medium-Term Strategy (Capacity Expansion)

Adopt the cost-minimizing expansion configuration:

- Open Izmir (with expansion)
- Open Samsun
- Open AnkNew

This configuration results in a total system cost of **\$23.56M**, including **\$12.0M fixed investment** (p.6). These locations align production with major demand regions and export flows, minimizing long-run transportation cost.

Long-Term Network Design Principles

- Prioritize geographic alignment with demand over minimizing fixed cost alone.
- Avoid expansions that do not generate sufficient transportation savings (e.g., Adana scenario increases cost to **\$27.06M**).

Risk Mitigation & Operational Flexibility

Operating at **90% capacity** increases total cost moderately to **\$25.51M (+8.3%)**, but improves resilience and maintenance flexibility (p.8). Maintaining capacity buffers may be justified under demand uncertainty and supply risk.

Financial Impact Summary

- Short-term savings: **~16.7% transportation reduction**
- Optimal long-term total cost: **\$23.56M**
- Cost penalty of poor location decisions: **+10–15%**

Appendix

A1. LP Model

optimize Shipment Plan

Amount of malt shipped from plant i to brewery j in year 1 (1000 ton)

	Istanbul	Ankara	Total	Capacity
Afyon	29.04	0.96	30.00	30.00
Konya	0	25.44	25.44	68.00
Import	0	0	0.00	20.00
Total malt	29.04	26.4		
Total beer	242	220	(Million lt)	

Amount of beer shipped from brewery j to distribution center k in year 1 (Million liters)

	Istanbul	Izmir	Antalya	Bursa	Kayseri	Samsun	Export (Izmir)	Total	constrain	Capacity
Istanbul	103	66	0	60	0	0	13	242	<=	250
Ankara	0	8	50	0	102	60	0	220.00	<=	220
Total	103	74	50	60	102	60	13			
Demands	103	74	50	60	102	60	13			
								Total shipping cost		11.80065
								(Million dollars)		

cost reduction 16.7%

B2. Sensitivity Analysis

Variable Cells

Cell	Name	Final Value	Reduced Cost	Objective Coefficient	Allowable Increase	Allowable Decrease
\$P\$47	Afyon Istanbul	29.04	0	0.0264374	0.0026438	0.000822429
\$Q\$47	Afyon Ankara	0.96	0	0.0166864	4.9E-05	0.0026438
\$P\$48	Konya Istanbul	0	0.010339	0.0368254	1E+30	0.010339
\$Q\$48	Konya Ankara	25.44	0	0.0167354	0.0026438	4.9E-05
\$P\$49	Import Istanbul	0	0.002884145	0.0317784	1E+30	0.002884145
\$Q\$49	Import Ankara	0	0.0142566	0.0325134	1E+30	0.0142566
\$P\$55	Istanbul Istanbul	103	0	0	0.031285714	0.003178368
\$Q\$55	Istanbul Izmir	66	0	0.040357143	0.013857143	0
\$R\$55	Istanbul Antalya	0	0.013857143	0.051714286	1E+30	0.013857143
\$S\$55	Istanbul Bursa	60	0	0.017357143	0.008857143	0.020535511
\$T\$55	Istanbul Kayseri	0	0.033428571	0.055142857	1E+30	0.033428571
\$U\$55	Istanbul Samsun	0	0.025071429	0.053	1E+30	0.025071429
\$V\$55	Istanbul Export (Izmir)	13	0	0.041857143	0	0.045035511
\$P\$56	Ankara Istanbul	0	0.031285714	0.032357143	1E+30	0.031285714
\$Q\$56	Ankara Izmir	8	0	0.041428571	0	0.013857143
\$R\$56	Ankara Antalya	50	0	0.038928571	0.013857143	0.041035511
\$S\$56	Ankara Bursa	0	0.008857143	0.027285714	1E+30	0.008857143
\$T\$56	Ankara Kayseri	102	0	0.022785714	0.033428571	0.024892654
\$U\$56	Ankara Samsun	60	0	0.029	0.025071429	0.031106939
\$V\$56	Ankara Export (Izmir)	0	0	0.042928571	1E+30	0

C3. Open/Expand

	efyon	konya	import(Izmir)	Beer Equivalent		efyon	konya	import(Izmir)	
Istanbul	30	0	0	250	Istanbul	0.026	0.037	0.032	
Ankara	0	21	0	175	Ankara	0.017	0.017	0.033	
Izmir	0	0	13.2	120	Izmir	0.02	0.031	0.004	
Sakarya	0	0	0	0	Sakarya	0.019	0.03	0.028	
Adana	0	6.66134E-16	0	5.55112E-15	Adana	0.032	0.022	0.048	
Samsun	0	10.8	0	90	Samsun	0.037	0.033	0.053	
IstNew	0	0	1.33227E-15	1.21115E-14	IstNew	0.026	0.037	0.032	
AnkNew	0	10.8	0	90	AnkNew	0.017	0.017	0.033	
total malt	30	42.6	13.2						malt cost 1.7298

Breweries	Istanbul	Izmir	Antalya	Bursa	Kayseri	Samsun	Export (Izmir)	sum	constrain
Istanbul	150.000	0.000	0.000	100.000	0.000	0.000	0.000	250.000	<= 250.000
Ankara	0.000	0.000	100.000	5.000	50.000	0.000	20.000	175.000	<= 220.000
Izmir	0.000	110.000	0.000	0.000	0.000	0.000	10.000	120.000	<= 120
Sakarya	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	<= 0
Adana	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	<= 0
Samsun	0.000	0.000	0.000	0.000	0.000	90.000	0.000	90.000	<= 90
IstNew	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	<= 0
AnkNew	0.000	0.000	0.000	0.000	90.000	0.000	0.000	90.000	<= 90
sum	150.000	110.000	100.000	105.000	140.000	90.000	30.000		
constrain									

(Million \$)	Yearly Equivalent		open (cost)		expend(cost)	max capacity
	Open(1/0)	Expand(1/0)				
Izmir	1	1	5	1		120
Sakarya	0	0	4	0.8		0
Adana	0	0	3.5	0.7		0
Samsun	1	0	2.5	0.5		90
IstNew	0	0	5.5	1.2		0
AnkNew	1	0	3.5	0.9		90

total fixed cost	12
total cost	23.55837143

beer cost	9.8285714
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D4. Alternative Scenario 1 - Binary variables (adana=1, samsun=0)

<i>(Million \$)</i>	<i>Yearly Equivalent</i>		open (cost)	expand(cost)	max capacity
	<i>Open(1/0)</i>	<i>Expand(1/0)</i>			
Izmir	1	1	5	1	120
Sakarya	0	0	4	0.8	0
Adana	1	0	3.5	0.7	90
Samsun	0	0	2.5	0.5	0
IstNew	0	0	5.5	1.2	0
AnkNew	1	0	3.5	0.9	90
total fixed cost	13				
total cost	27.05792857				

E5. Alternative Scenario 2 - No Izmir

(Million \$)	Yearly Equivalent		open (cost)	expand(cost)	max capacity
	Open(1/0)	Expand(1/0)			
Izmir	0	0	5	1	0
Sakarya	1	0	4	0.8	90
Adana	0	0	3.5	0.7	0
Samsun	1	0	2.5	0.5	90
stNew	0	0	5.5	1.2	0
AnkNew	1	0	3.5	0.9	90
total fixed cost	10				
total cost	26.08612857				

F6. 90% Capacity

<i>Breweries</i>	Istanbul	Izmir	Antalya	Bursa	Kayseri	Samsun	Export (Izmir)	sum	constrain
Istanbul	150.000	2.000	0.000	69.000	0.000	0.000	0.000	221.000	<= 225.000
Ankara	0.000	0.000	58.000	0.000	140.000	0.000	0.000	198.000	<= 198.000
Izmir	0.000	108.000	0.000	0.000	0.000	0.000	0.000	108.000	<= 108
Sakarya	0.000	0.000	42.000	36.000	0.000	0.000	30.000	108.000	<= 108
Adana	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	<= 0
Samsun	0.000	0.000	0.000	0.000	0.000	90.000	0.000	90.000	<= 108
IstNew	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	<= 0
AnkNew	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	<= 0
sum	150.000	110.000	100.000	105.000	140.000	90.000	30.000		
constrain									

<i>(Million \$)</i>	<i>Yearly Equivalent</i>				
	<i>Open(1/0)</i>	<i>Expand(1/0)</i>	open (cost)	expand(cost)	max capacity
Izmir	1	1	5	1	108
Sakarya	1	1	4	0.8	108
Adana	0	0	3.5	0.7	0
Samsun	1	1	2.5	0.5	108
IstNew	0	0	5.5	1.2	0
AnkNew	0	0	3.5	0.9	0

total fixed cost	13.8
total cost	25.5140387